

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024****Course: B. Tech Branch: Electronics and Computer Engineering/ Electronics and****Computer Science****Semester: V****Subject Code & Name: (BTEPC502) & (Digital Signal & Image Processing)****Max Marks: 60****Date: 08/02/2025****Duration: 3 Hr.****Instructions to the Students:**

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The value of continuous impulse function $\delta(t)$ at $t \neq 0$	L2 / CO1	1
	a. 0 b. 1 c. ∞ d. None		
2	All periodic signals are:	L2 / CO1	1
	a. Power Signals b. Energy Signals c. Neither energy nor power d. None		
3	The system is said to be linear if it follows the principles:	L2 / CO1	1
	a. Homogeneity b. Superposition c. Both a & b d. None		
4	Identify no. of multiplications to be performed in order to calculate N-point DFT:	L3 / CO3	1
	a. N^2 b. (N^2-N) c. $\log N$ d. $N * \log(N)$		
5	ROC of a Z-transform of discrete unit step causal signal is represented as:	L3 / CO2	1
	a. $ z > 1$ b. $ z < 1$ c. $1 < z < 2$ d. $ z > 2$		
6	Identify the morphological operation performed on image	L3 / CO5	1
	a. Dilation b. Erosion c. Boundary extraction d. All of the above mentioned		
7	Identify the true statement related to Convolution and Correlation operations on a given image:	L3 / CO4	1
	a. Image is pre-rotated by 180° for correlation b. Image is pre-rotated by 180° for convolution c. Image is pre-rotated by 90° for correlation d. Image is pre-rotated by 90° for convolution		
8	Histogram Matching is also referred as:	L2 / CO4	1
	a. Histogram equalisation b. Histogram specification		

	c. Histogram linearisation d. None of the mentioned		
9	Piecewise Linear Transformation function involves which of the following? a. Bit-plane slicing b. Intensity level slicing c. Contrast stretching d. All of the mentioned	L2 / CO3	1
10	Segmentation is a process of a. Low level b. Edge level c. Mid-level d. High level	L2 / CO3	1
11	A Spatial Low pass filter whose all coefficients are equal is referred: a. Median Filter b. Box or Mean Filter c. Max Filter d. Min Filter	L2 / CO3	1
12	The step which extracts image components useful in the representation and description of shape of object referred as: a. Segmentation b. Representation & description c. Compression d. Morphological processing	L2 / CO3	1
Q. 2	Solve the following.		12
A)	Compare Analog Systems with Digital systems in terms of Bandwidth limitation, System design, Accuracy, Storage and processing of low frequency signal.	L3 / CO1	6
B)	Identify whether following signals are periodic or non-periodic. Specify its fundamental period (if signal is periodic in nature) 1. $x(n) = e^{j6\pi n}$ 2. $x(n) = \cos\left(\frac{2\pi n}{3}\right)$	L3 / CO1	6
Q.3	Solve the following.		12
A)	Illustrate the following properties of Z-transform with valid justification. 1. Linearity Property 2. Time Shifting Property 3. Differentiation Property	L2 / CO2	6
B)	Find the output of the Discrete system if the system is having the input $x(n) = \{4, 4, 4\}$ and system's impulse response $h(n) = \{2, 2\}$. Use N-point DFT & IDFT.	L2 / CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Explain the various fundamental steps in Digital Image Processing in details	L2 / CO3	6
B)	Find whether the mentioned two pixels P & Q in a given Image (I) are: 1. 4-Adjacent 2. 8-Adjacent 3. m-Adjacent	L2 / CO3	6

	Assume: $V=\{1,2,3\}$	<table><tr><td>5</td><td>6</td><td>8</td><td>4</td><td>3(Q)</td></tr><tr><td>4</td><td>9</td><td>1</td><td>2(P)</td><td>4</td></tr><tr><td>5</td><td>2</td><td>6</td><td>2</td><td>1</td></tr></table>	5	6	8	4	3(Q)	4	9	1	2(P)	4	5	2	6	2	1						
5	6	8	4	3(Q)																			
4	9	1	2(P)	4																			
5	2	6	2	1																			
C)	Identify the Euclidean Distance (D_e), City-block Distance (D_c) and Chess-board Distance (D_b) between the pixels P & Q of a given Grayscale Image $I(x, y)$ <table><tr><td>5</td><td>9</td><td>8</td><td>7</td><td>6</td></tr><tr><td>0</td><td>1(P)</td><td>6</td><td>2</td><td>4</td></tr><tr><td>6</td><td>8</td><td>7</td><td>2</td><td>4</td></tr><tr><td>3</td><td>3</td><td>2</td><td>6</td><td>5(Q)</td></tr></table> Specify the result as (D_e, D_c, D_b)	5	9	8	7	6	0	1(P)	6	2	4	6	8	7	2	4	3	3	2	6	5(Q)	L3 / CO3	6
5	9	8	7	6																			
0	1(P)	6	2	4																			
6	8	7	2	4																			
3	3	2	6	5(Q)																			
Q.5	Solve Any Two of the following.			12																			
A)	Apply the grey level transformation on the given image having the intensity range (42 – 115) into transformed image having the intensity range (162 – 224). $F(x, y) = \begin{bmatrix} 45 & 87 & 65 & 75 \\ 48 & 109 & 114 & 89 \\ 93 & 68 & 98 & 100 \end{bmatrix}$ Specify the result of operation.	L2 / CO4	6																				
B)	Consider a 3-bit image ($L=8$) of size 8×8 pixels with the histogram specified in the given table. Perform Histogram Equalization. <table><tr><td>r_k</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>h_k</td><td>8</td><td>10</td><td>10</td><td>2</td><td>12</td><td>16</td><td>4</td><td>2</td></tr></table> Specify the result of operation.	r_k	0	1	2	3	4	5	6	7	h_k	8	10	10	2	12	16	4	2	L3 / CO4	6		
r_k	0	1	2	3	4	5	6	7															
h_k	8	10	10	2	12	16	4	2															
C)	Explain the concept of Spatial filtering and illustrate the following Spatial Low pass filters with suitable examples: 1. Mean Filters 2. Weighted average Filters 3. Median Filters	L2 / CO4	6																				
Q. 6	Solve Any Two of the following.			12																			
A)	Apply the following Morphological operations on the given image. 1. Opening 2. Closing Image = $\begin{bmatrix} 0111 \\ 1111 \\ 0111 \\ 1111 \end{bmatrix}$ & Structuring Element = $\begin{bmatrix} 11 \\ 11 \end{bmatrix}$ Specify the result of operations.	L3 / CO5	6																				
B)	Explain Splitting and Merging image segmentation algorithms in detail.	L2 / CO3	6																				
C)	Illustrate the following morphological operations with suitable example 1. Hit or miss transformation 2. Boundary Extraction	L2 / CO3	6																				
	*** End ***																						